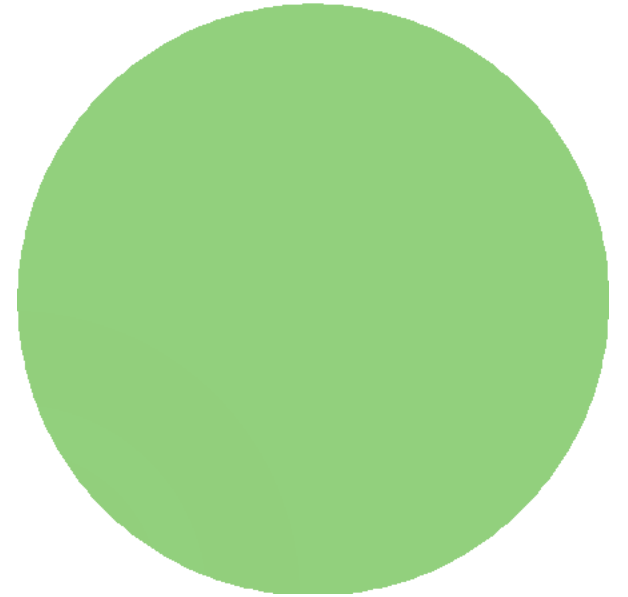


Probability Distribution



Contents

- ▶ Probability Distribution
- ▶ Discrete and Continuous Distribution
- ▶ Mathematical Expectations
- ▶ Binomial Distribution
- ▶ Poisson Distribution
- ▶ Uniform Distribution
- ▶ Exponential Distribution
- ▶ Normal Distribution, Normal Curve and Its Characteristics
- ▶ Area Under The Normal Curve

Random Variable

A random variable is a variable that takes on **numerical values** as a result of a random experiment or measurement; associates a numerical value with each possible outcome.

The differences between variable and random variable are-

- Random variable always takes **numerical values**
- There is a **probability associated** with each possible values

Random variable is denoted by capital letters such as X, Y, Z etc.

And the possible outcomes are denoted by small letters such as x, y, z etc.

Random Variable

4

Example 1:

A coin is tossed. It has two possible outcomes- Head and Tail.

Consider a variable, $X = \text{outcome of a coin toss} = \begin{cases} H, & \text{if Head appears} \\ T, & \text{if Tail appears} \end{cases}$

Here, $S = \{H, T\}$.

But, these are not numerical values.

Random Variable

5

Example 1(contd.):

Consider a variable, X = Number of heads obtained in a trial

Then, $X = \begin{cases} 1, & \text{if Head appears} \\ 0, & \text{if Tail appears} \end{cases}$

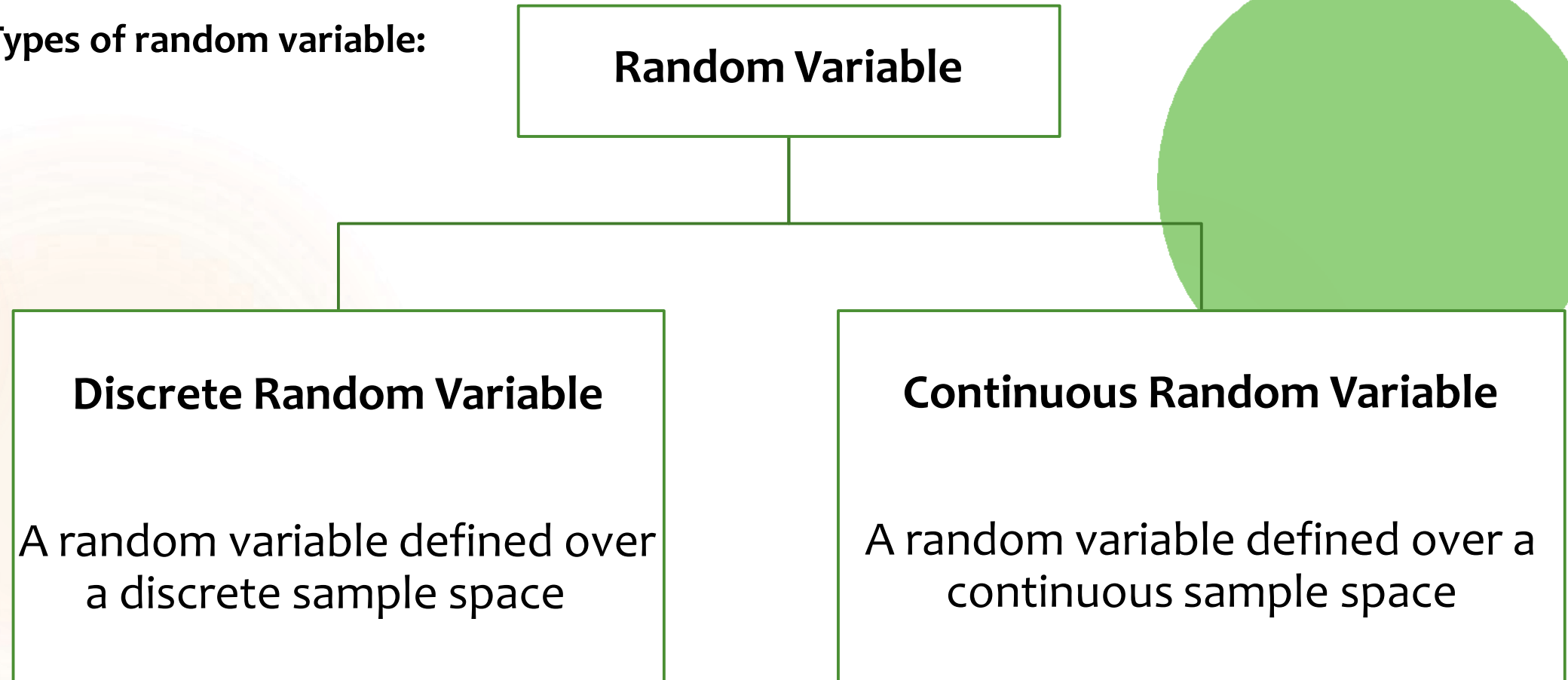
For a fair coin, we can write, $P(X=1) = \frac{1}{2}$ and $P(X=0) = \frac{1}{2}$

So, X is a random variable.

Random Variable

6

Types of random variable:



Random Variable

7

Examples:

Discrete Random Variable:

1. X = Number of correct answers in a 100-MCQ test = $0, 1, 2, \dots, 100$
2. X = Number of cars passing a toll both in a day = $0, 1, 2, \dots, \infty$
3. X = Number of balls required to take the first wicket = $1, 2, 3, \dots, \infty$

Continuous Random Variable:

1. X = Weight of a person. $0 < X < \infty$
2. X = Monthly Profit. $-\infty < X < \infty$

Probability Distributions

8

Distribution of the probabilities among the different values of a random variable.

Discrete probability distribution- probability distribution of a discrete random variable

Continuous probability distribution- probability distribution of a continuous random variable

Probability Distributions

Examples:

Discrete probability distribution-

- Tossing a coin 2 times.

X = Number of Heads appeared

$S = \{HH, HT, TH, TT\}$

x	0	1	2
$P(x)$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

Probability Distributions

10

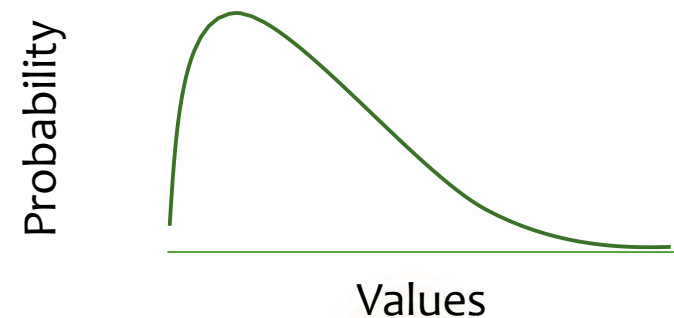
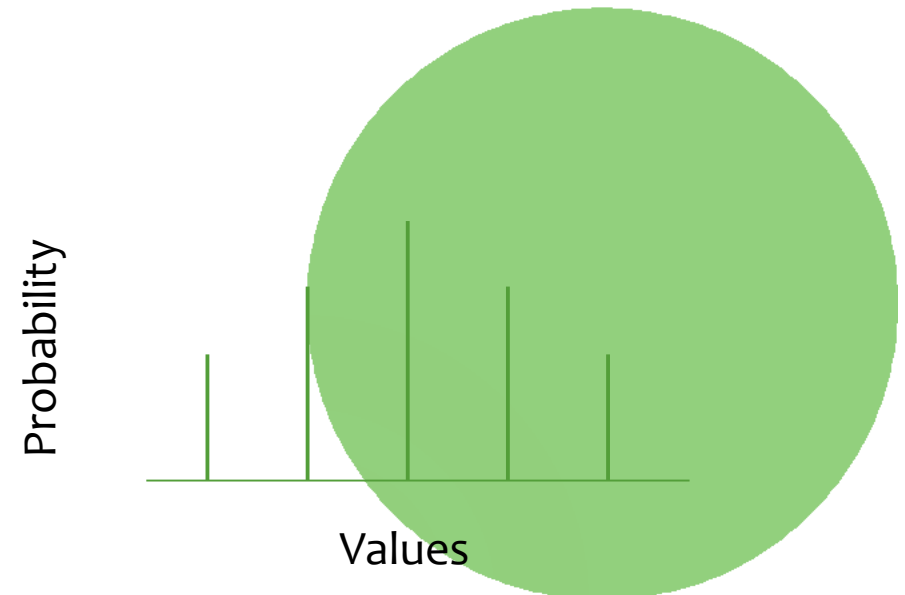
Different types of probability distributions:

Discrete probability distribution-

1. Bernoulli Distribution
2. *Binomial Distribution*
3. Poisson Distribution etc.

Continuous probability distribution-

1. Uniform Distribution
2. *Normal Distribution*
3. Exponential Distribution
4. t-distribution etc.



PMF and PDF

11

Probability Mass Function (pmf)- the probability distribution function of a discrete random variable X is called a pmf and is denoted by $p(x)$

Probability Density Function (pdf)- the probability distribution function of a continuous random variable X is called a pdf and is denoted by $f(x)$

Mathematical Expectations

12

- For a discrete random variable X with pmf $p(x)$, the mathematical expectation of X is-

$$\mu = E(X) = \sum_x x p(x)$$

- For a continuous random variable X with pdf $f(x)$, the mathematical expectation of X is-

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

Mathematical expectation is also known as population mean or expected value.

Mathematical Expectations

13

$$E(X^2) = \begin{cases} \sum_x x^2 p(x) & , \text{if } x \text{ is a discrete r.v.} \\ \int x^2 f(x) dx & , \text{if } x \text{ is a continuous r.v.} \end{cases}$$

Variance:

$$\sigma^2 = \text{Var}(X) = E[X - E(X)]^2 = E(X^2) - [E(X)]^2 = E(X^2) - \mu^2$$

Standard deviation: $\sigma = \sqrt{\text{Var}(X)}$

Properties of Mathematical Expectations

14

Let, c is a constant number

X and Y are two independent random variables

1. $E(c) = c$
2. $E(c X) = c E(x)$
3. $E(X + c) = E(x) + c$
4. $E(X+Y) = E(X) + E(Y)$
5. $E(X-Y) = E(X) - E(Y)$
6. $E(XY) = E(X) \cdot E(Y)$

1. $\text{Var}(c) = 0$
2. $\text{Var}(c X) = c^2 \text{Var}(x)$
3. $\text{Var}(X + c) = \text{Var}(x)$
4. $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$
5. $\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y)$

Mathematical Expectation

15

Example 2-

A company estimates the net profit on a new product, it is launching, to be Rs. 3 million during first year, if it is 'successful', Rs. 1 million if it is 'moderately successful', and a loss of Rs. 1 million if it is 'unsuccessful'.

The company assigns the following probabilities to first year prospects for the product-

Successful: 0.25, Moderately successful: 0.40, and Unsuccessful: 0.35

What are the **expected value** and **standard deviation** of the first year net profit for the product? Also, find the expected value of net profit if there is a fixed cost of Rs. 0.2 million, whatever the success status is.

Mathematical Expectation

16

Solution-

Let,

X = Net profit on the new product in the 1st year (Rs. Million)

Given that,

x	3	1	-1
$P(x)$	0.25	0.4	0.35

$$\begin{aligned}\text{Expected net profit, } E(X) &= \sum x p(x) = (3 * 0.25) + (1 * 0.4) + (-1 * 0.35) \\ &= 0.8 \text{ million}\end{aligned}$$

Mathematical Expectation

17

Solution (contd.)-

$$\begin{aligned} E(X^2) &= \sum x^2 p(x) = (3^2 * 0.25) + (1^2 * 0.4) + ((-1)^2 * 0.35) \\ &= (9 * 0.25) + (1 * 0.4) + (1 * 0.35) = 3 \end{aligned}$$

$$\begin{aligned} Var(X) &= E(X^2) - [E(X)]^2 = 3 - 0.8^2 = 2.36 \\ \therefore SD(X) &= \sqrt{Var(X)} = \sqrt{2.36} = 1.54 \text{ million} \end{aligned}$$

If there is a fixed cost of Rs. 0.2 million, then expected net profit-

$$E(X - 0.2) = E(X) - 0.2 = 0.8 - 0.2 = 0.6 \text{ million}$$

Binomial Distribution

18

Bernoulli trial:

A trial that has only two possible outcomes (often called 'Success' and 'Failure')

Outcome	Success	failure
Probability	p	$1-p$

Let,

- n independent Bernoulli trials are performed
- Each trial has the same probability of success, p

Binomial Distribution

19

Let,

X= number of success in n trials

Then, X is a binomial random variable with distribution function (pmf),

$$p(x) = {}^nC_x p^x (1-p)^{n-x} \quad ; x = 0, 1, 2, \dots, n$$
$$= \frac{n!}{(n-x)! x!} p^x (1-p)^{n-x}$$

Here, $n! = n(n-1)(n-2) \dots 1$ $0! = 1$ $1! = 1$ $2! = 2 \times 1 = 2$ $3! = 3 \times 2 \times 1 = 6$

We write it as, $X \sim \text{binomial}(n, p)$

Binomial Distribution

20

Mean of the binomial distribution, $\mu = E(X) = \sum x p(x) = np$

Variance of binomial distribution, $\sigma^2 = E(X^2) - \mu^2 = np(1 - p) = npq$

Standard deviation of binomial distribution, $\sigma = \sqrt{npq}$

Binomial Distribution

21

Example 3:

There are 3 multiple choice questions in a MCQ test. Each MCQ consists of four possible choices and only one of them is correct. If an examinee answers those MCQ randomly (without knowing the correct answers)

- a. What is the probability that exactly any two of the answers will be correct?
- b. What is the probability that at least two of the answers will be correct?
- c. What is the probability that at most two of the answers will be correct?
- d. What will be the average or expected number of correct answers?
- e. Also, find the standard deviation of number of correct answers.

Binomial Distribution

22

Solution:

Let,

X= number of correct answers selected in 3 MCQs

Here, p = probability of selecting correct answer per question = $\frac{1}{4} = 0.25$

$$\therefore X \sim \text{binomial } (n = 2, p = 0.25)$$

$$\begin{aligned} p(x) &= {}^3C_x (0.25)^x (1 - 0.25)^{3-x} && ; x = 0, 1, 2, 3 \\ &= \frac{3!}{(3-x)! x!} (0.25)^x (0.75)^{3-x} \end{aligned}$$

Binomial Distribution

23

Solution (contd.):

a. probability that exactly any two of the answers will be correct-

$$\begin{aligned} P(X = 2) &= \frac{3!}{(3-2)! 2!} (0.25)^2 (0.75)^{3-2} \\ &= \frac{3!}{1! 2!} (0.25)^2 (0.75)^1 = \frac{3 * 2 * 1}{1 * (2 * 1)} * 0.0625 * 0.75 = 0.141 \end{aligned}$$

b. probability that at least two of the answers will be correct-

$$\begin{aligned} P(X \geq 2) &= P(X = 2) + P(X = 3) \\ &= \frac{3!}{(3-2)! 2!} (0.25)^2 (0.75)^{3-2} + \frac{3!}{(3-3)! 3!} (0.25)^3 (0.75)^{3-3} \\ &= \frac{3!}{1! 2!} (0.25)^2 (0.75)^1 + \frac{3!}{0! 3!} (0.25)^3 (0.75)^0 = 0.141 + 0.016 = 0.157 \end{aligned}$$

Binomial Distribution

24

Solution (contd.):

c. probability that at most two of the answers will be correct-

$$\begin{aligned} P(X \leq 2) &= P(X = 0) + P(X = 1) + P(X = 2) \\ &= \frac{3!}{(3-0)! 0!} (0.25)^0 (0.75)^{3-0} + \frac{3!}{(3-1)! 1!} (0.25)^1 (0.75)^{3-1} \\ &\quad + \frac{3!}{(3-2)! 2!} (0.25)^2 (0.75)^{3-2} \\ &= \frac{3!}{3! 0!} (0.25)^0 (0.75)^3 + \frac{3!}{2! 1!} (0.25)^1 (0.75)^2 + \frac{3!}{1! 2!} (0.25)^2 (0.75)^1 \\ &= 0.422 + 0.422 + 0.141 = 0.985 \end{aligned}$$

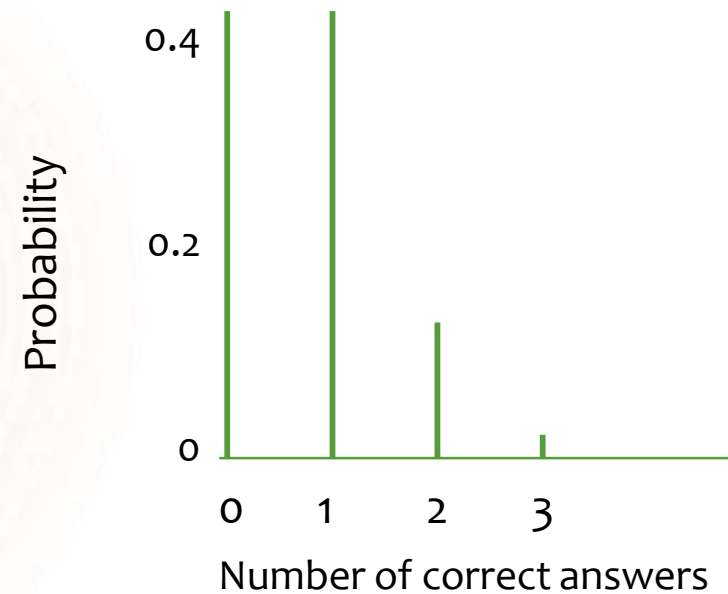
d. $E(X) = np = 3 * .25 = 0.75$

e. $SD(X) = \sqrt{npq} = \sqrt{3 * 0.25 * 0.75} = 0.75$

Binomial Distribution

25

X	0	1	2	3
P(x)	0.422	0.422	0.141	0.016



Poisson Distribution

Let,

X = a random variable usually counts or number of occurrences

Then, X is a Poisson random variable with distribution function (pmf),

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad ; x = 0, 1, 2, \dots$$

We write it as, $X \sim \text{Poisson}(\lambda)$

Poisson Distribution

27

Mean of the Poisson distribution, $\mu = E(X) = \sum x p(x) = \lambda$

Variance of Poisson distribution, $\sigma^2 = E(X^2) - \mu^2 = \lambda$

Standard deviation of Poisson distribution, $\sigma = \sqrt{\lambda}$

Poisson Distribution

28

Example 4:

The average number of errors on a page of a certain magazine is 0.2.
What is the probability that the next page (or a randomly selected page) you read contains

- i. 0 (zero) error?
- ii. 2 or more errors?
- iii. What is the average error per page?
- iv. Also, find standard deviation of the number of errors.

Poisson Distribution

29

Solution:

Let,

X = number of errors in a page

Here, λ = average number of errors per page = 0.2

$$\therefore X \sim \text{Poisson} (\lambda = 0.2)$$

$$\begin{aligned} p(x) &= \frac{e^{-\lambda} \lambda^x}{x!} \quad ; x = 0, 1, 2, \dots \\ &= \frac{e^{-0.2} 0.2^x}{x!} \end{aligned}$$

Poisson Distribution

30

Solution:

i.
$$P(X = 0) = \frac{e^{-0.2} 0.2^x}{x!} = \frac{e^{-0.2} 0.2^0}{0!} = \frac{e^{-0.2} * 1}{1} = 0.8187$$

ii.
$$\begin{aligned} P(X \geq 2) &= 1 - P(X < 2) = 1 - [P(X = 0) + P(X = 1)] \\ &= \frac{e^{-0.2} 0.2^0}{0!} + \frac{e^{-0.2} 0.2^1}{1!} = 1 - [e^{-0.2} + e^{-0.2} * 0.2] = 0.01756 \end{aligned}$$

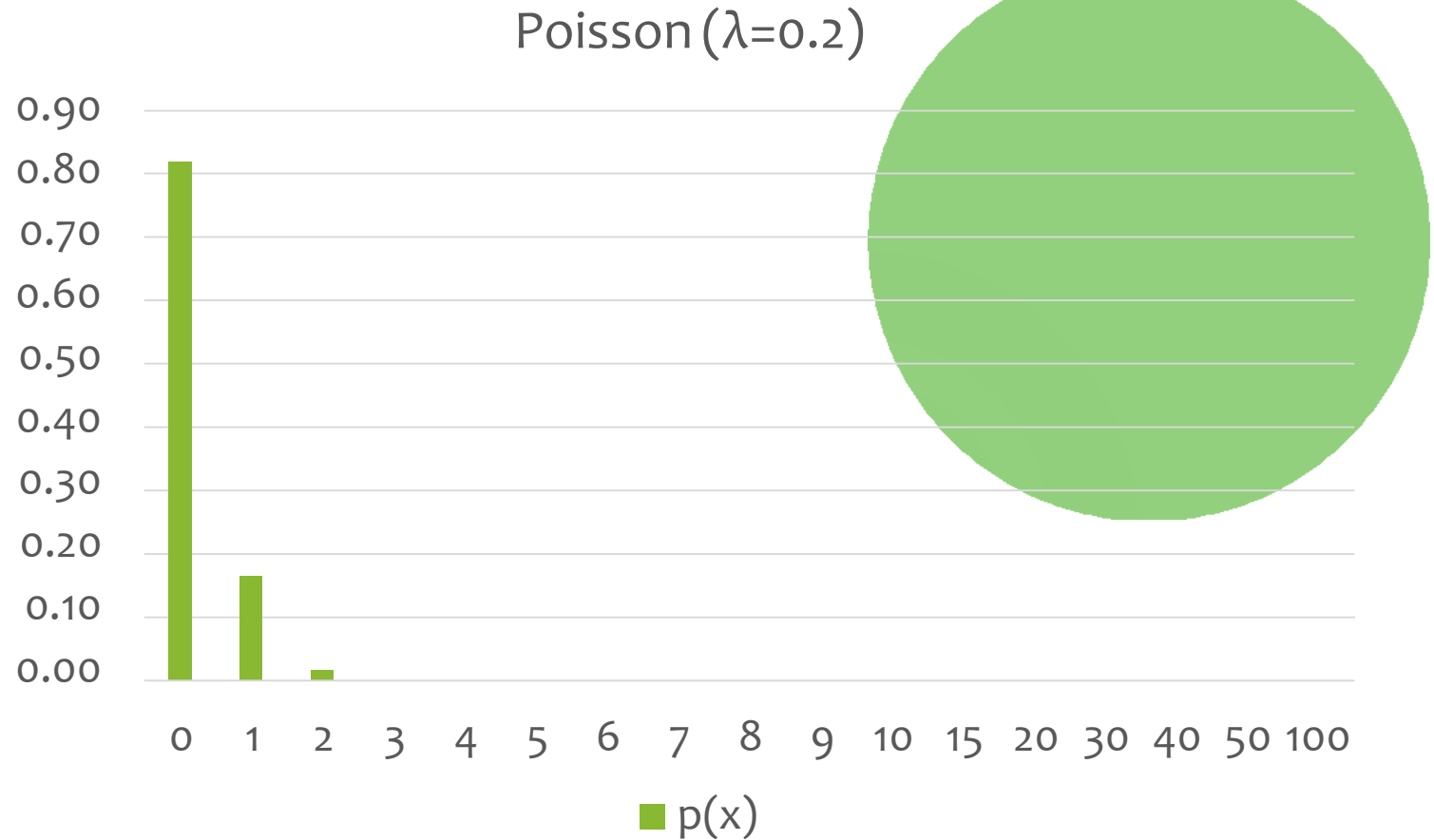
iii. Average number of errors, $E(X) = \lambda = 0.2$

iv. Standard deviation, $SD(X) = \sqrt{\lambda} = \sqrt{0.2} = 0.45$

Poisson Distribution

31

x	p(x)
0	0.82
1	0.16
2	0.02
3	0.00
4	0.00
5	0.00
6	0.00
7	0.00
8	0.00
9	0.00
10	0.00
15	0.00
20	0.00
30	0.00
40	0.00
50	0.00
100	0.00



Uniform Distribution

32

Let,

X is a continuous random variable

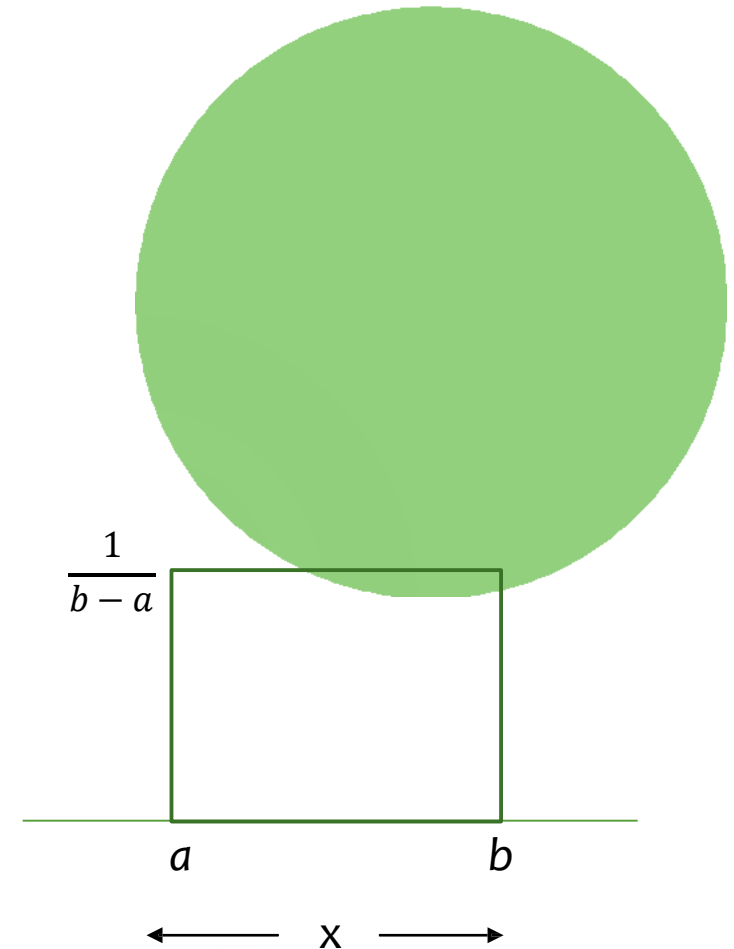
Then, if X has a probability density function (pdf),

$$f(x) = \frac{1}{b-a} \quad ; \quad a < x < b$$

We write it as, $X \sim \text{uniform}(a, b)$

$$\text{Mean, } E(X) = \frac{a+b}{2}$$

$$\text{Variance, } V(X) = \frac{(b-a)^2}{12}$$



Uniform Distribution

33

Example 5:

The waiting time (in minutes) for train is uniform (10, 50).

Find-

- a. The probability that you have to wait at least 20 minutes.
- b. Average waiting time.
- c. Standard deviation of waiting time.

Uniform Distribution

34

Solution:

Let,

X= waiting time (in minutes)

$\therefore X \sim \text{uniform}(10, 50)$

$$\begin{aligned} f(x) &= \frac{1}{b-a} \quad ; \quad a < x < b \\ &= \frac{1}{50-10} ; 10 < x < 50 \\ &= \frac{1}{40} \end{aligned}$$

Uniform Distribution

35

Solution:

- a. Probability that you have to wait at least 20 minutes-

$$\begin{aligned} P[X \geq 20] &= \int_{20}^{50} f(x) dx = \int_{20}^{50} \frac{1}{40} dx = \frac{1}{40} \int_{20}^{50} 1 dx \\ &= \frac{1}{40} [x]_{20}^{50} = \frac{1}{40} [50 - 20] = \frac{30}{40} = \frac{3}{4} = 0.75 \end{aligned}$$

- b. Average waiting time-

$$E(X) = \frac{a + b}{2} = \frac{10 + 50}{2} = \frac{60}{2} = 30 \text{ minutes}$$

Uniform Distribution

36

Solution:

c. Standard deviation of waiting time-

$$SD(X) = \sqrt{\frac{(b - a)^2}{12}} = \sqrt{\frac{(50 - 10)^2}{12}} = \sqrt{\frac{(40)^2}{12}} = 3.65 \text{ minutes}$$

Exponential Distribution

37

A continuous random variable X is said to follow exponential distribution, if its pdf is,

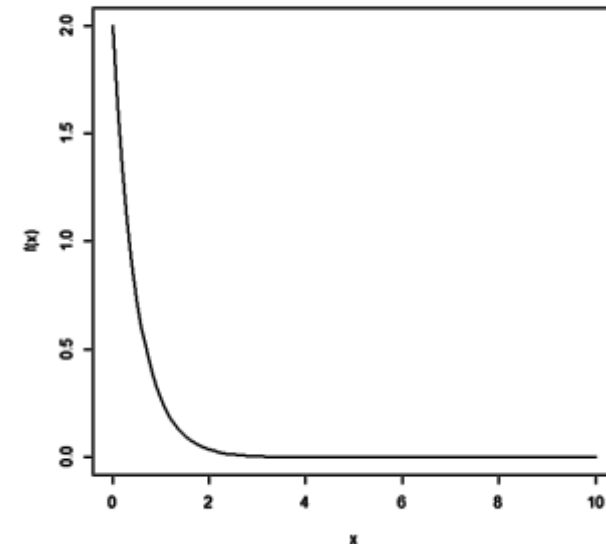
$$f(x) = \lambda e^{-\lambda x}; \quad x > 0, \text{ where } \lambda > 0$$

We write as, $X \sim \exp(\lambda)$

Here, X is usually time until certain event occurs.

$$\text{Mean, } E(X) = \frac{1}{\lambda}$$

$$\text{Variance, } V(X) = \frac{1}{\lambda^2}$$



Exponential Distribution

38

Example 6:

Average time required to repair a machine is 0.5 hours. What is the probability that the next repair will take more than 2 hours?

Solution:

Let, X = time required to repair the machine

$$\therefore X \sim \exp(\lambda = 2) \quad \left[\text{since, } E(X) = \frac{1}{\lambda} = 0.5 \Rightarrow \lambda = \frac{1}{0.5} = 2 \right]$$

$$\begin{aligned} \Pr[X > 2] &= \int_2^{\infty} \lambda e^{-\lambda x} dx = \left[-e^{-\lambda x} \right]_2^{\infty} = \left[-0 + e^{-2\lambda} \right] = e^{-2\lambda} = e^{-2 \times 2} \\ &= e^{-4} = 0.0183 \end{aligned}$$

Normal Distribution

39

Let,

X is a continuous random variable

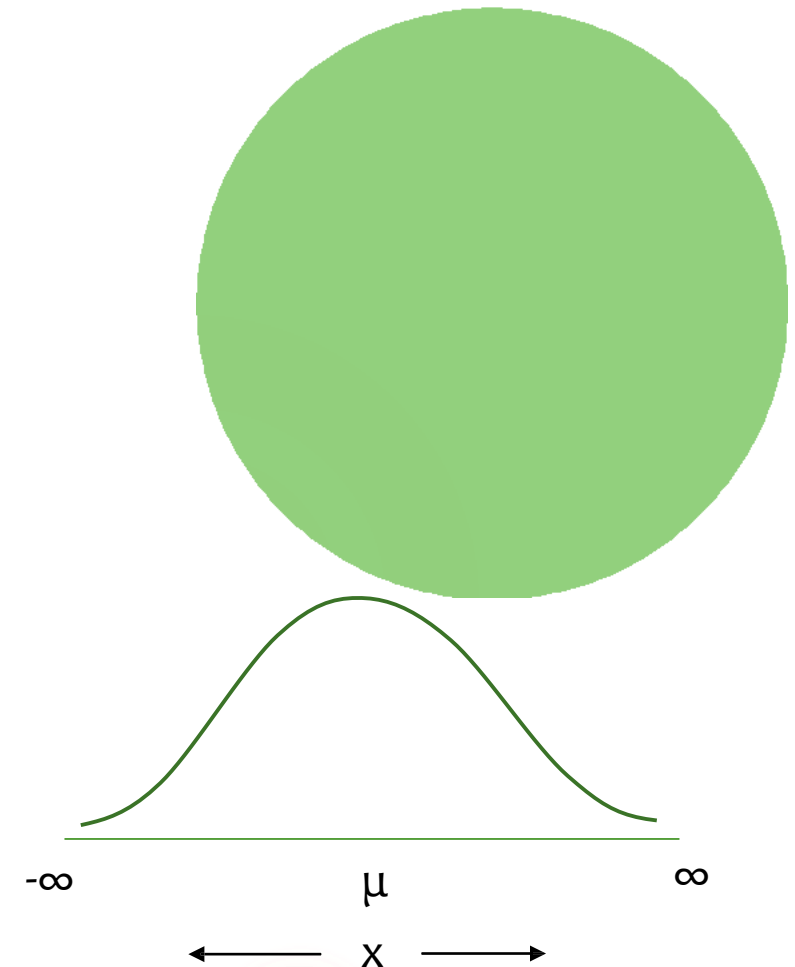
Then, if X has a probability density function (pdf),

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} ; -\infty < x < \infty$$

We write it as, $X \sim N(\mu, \sigma^2)$

Mean, $E(X) = \mu$

Variance, $V(X) = \sigma^2$



Standard Normal Distribution

40

Let,

$$Z = \frac{X - \mu}{\sigma}$$

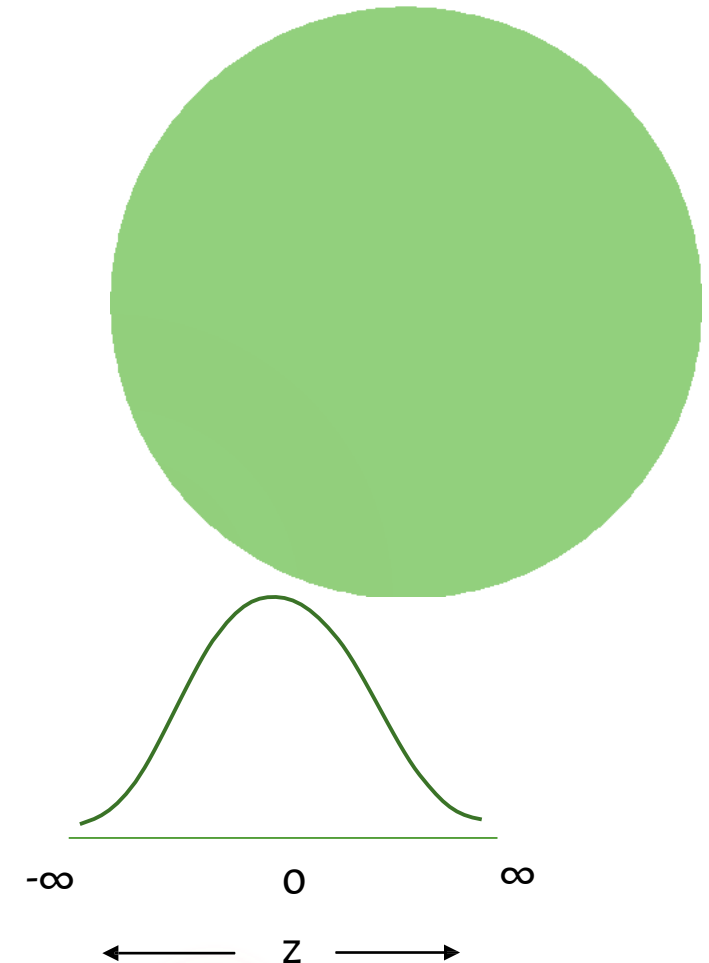
Then, Mean, $E(Z) = 0$

Variance, $V(Z) = 1$

And, if Z has a probability density function (pdf),

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} ; -\infty < z < \infty$$

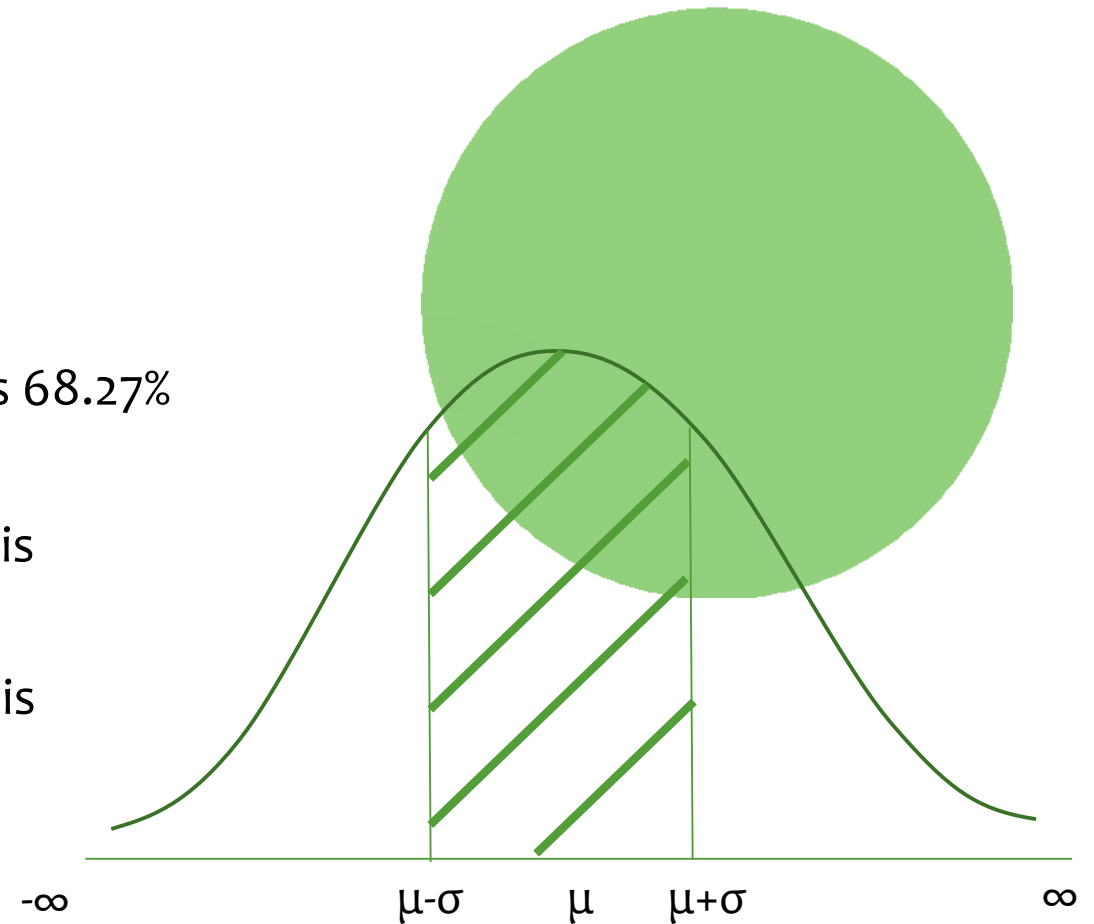
We write it as, $Z \sim N(0, 1)$



Characteristics of a Normal Distribution

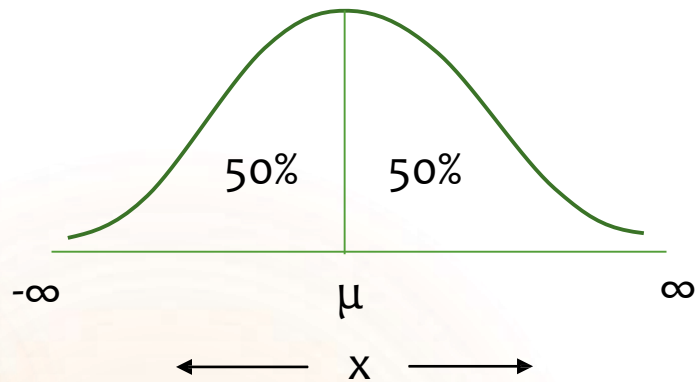
41

1. Mean= Median = Mode
2. Symmetric and Mesokurtic
3. Bell-shaped curve
4. The area under the curve lying between $\mu \pm \sigma$ is 68.27% of the total area
5. The area under the curve lying between $\mu \pm 2\sigma$ is 95.45% of the total area
6. The area under the curve lying between $\mu \pm 3\sigma$ is 99.73% of the total area



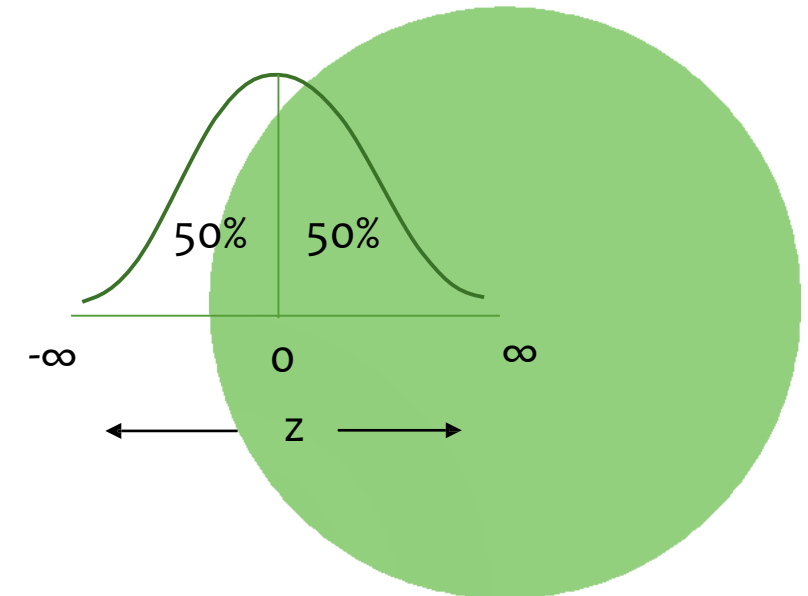
Characteristics of a Normal Distribution

42



$$P[X < \mu] = P[X > \mu] = 0.5$$

$$P[X < -x] = P[X > x]$$



$$P[Z < 0] = P[Z > 0] = 0.5$$

$$P[Z < -z] = P[Z > z]$$

Normal Distribution Table

Z-table

- Normal distribution table provides probabilities for $N(0,1)$ i.e. for standard normal distribution
- Usually, normal table gives $P[0 < Z < z]$ for positive values of Z .
- For other values, we can use the property of symmetry with median 0 of standard normal distribution
- To find probabilities for a normal random variable X , we can transform the probability statement about X in terms of probability statement for Z and then calculate the probability using the standard normal distribution table or Z-table

$$P[X < a] = P\left[\frac{X - \mu}{\sigma} < \frac{a - \mu}{\sigma}\right] = P\left[Z < \frac{a - \mu}{\sigma}\right]$$

Finding Area Under the Normal Curve using Z-table

Example 7:

The number of viewers of a TV show per week has a mean of 29 million with a standard deviation of 5 million. Assume that, the number of viewers of that show follows a normal distribution.

What is the probability that, next week's show will-

- a. Have between 30 and 34 million viewers?
- b. Have at least 23 million viewers?
- c. Exceed 40 million viewers?

Finding Area Under the Normal Curve using Z-table

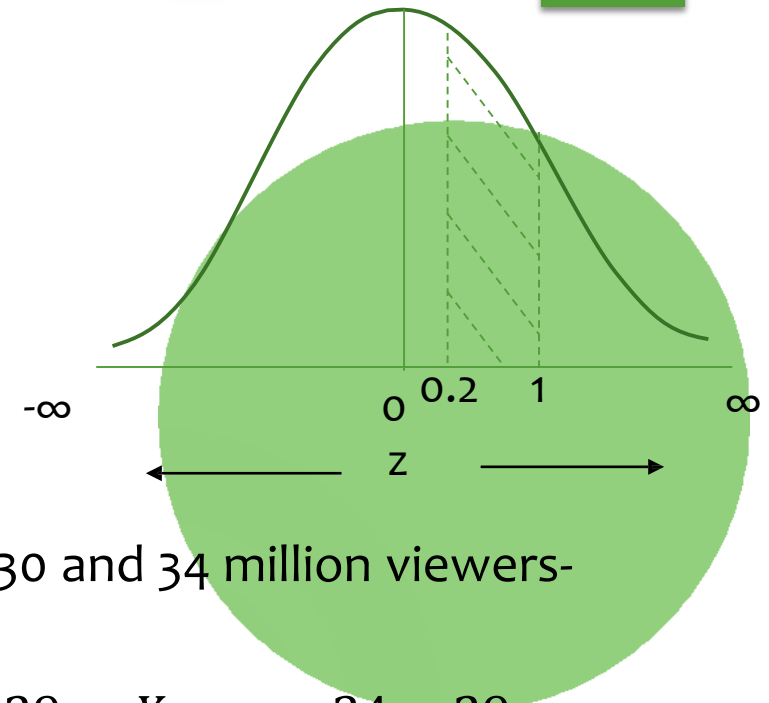
Solution:

Let, X = Number of viewers of the show per week (in million)

$$\therefore X \sim N(\mu, \sigma^2)$$

- a. the probability that, next week's show will have between 30 and 34 million viewers-

$$\begin{aligned} P[30 \leq X \leq 34] &= P\left[\frac{30 - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{34 - \mu}{\sigma}\right] = P\left[\frac{30 - 29}{5} \leq \frac{X - \mu}{\sigma} \leq \frac{34 - 29}{5}\right] \\ &= P[0.20 \leq Z \leq 1] = P[0 \leq Z \leq 1] - P[0 \leq Z \leq 0.2] = 0.3413 - 0.0793 \\ &= 0.262 \end{aligned}$$



Finding Area Under the Normal Curve using Z-table

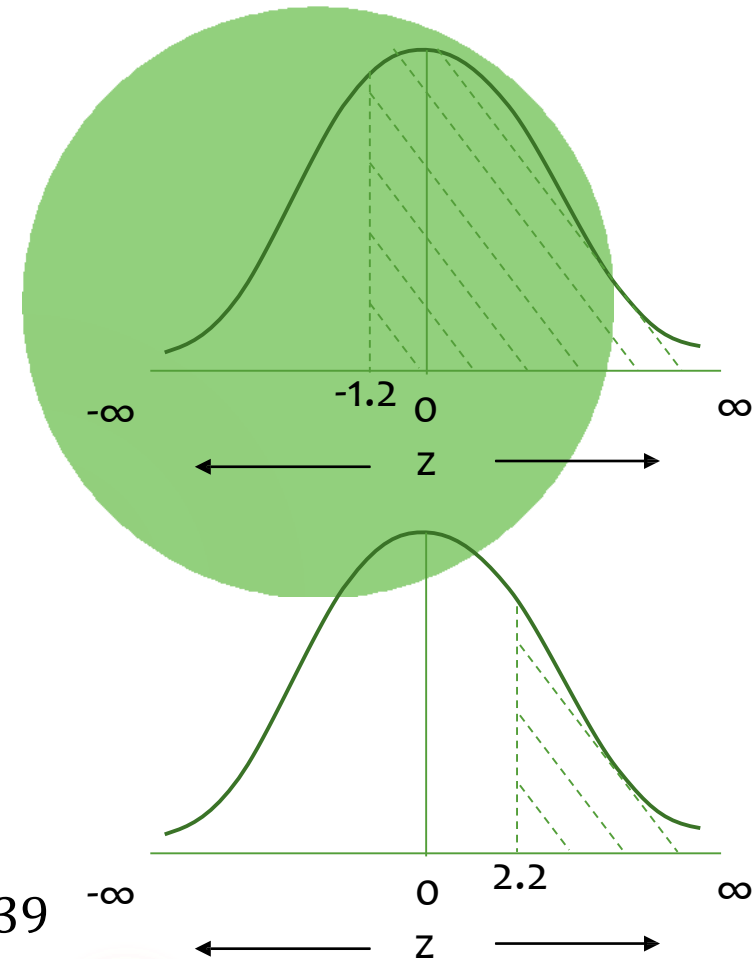
Solution (contd.):

b. the probability that, next week's show will have at least 23 million viewers-

$$\begin{aligned} P[X \geq 23] &= P\left[\frac{X - \mu}{\sigma} \geq \frac{23 - \mu}{\sigma}\right] = P\left[\frac{X - \mu}{\sigma} \geq \frac{23 - 29}{5}\right] \\ &= P[Z \geq -1.2] = P[-1.2 \leq Z \leq 0] + P[Z \geq 0] = 0.3849 + 0.5 \\ &= 0.8849 \end{aligned}$$

c. the probability that, next week's show will exceed 40 million viewers-

$$\begin{aligned} P[X > 40] &= P\left[\frac{X - \mu}{\sigma} > \frac{40 - \mu}{\sigma}\right] = P\left[\frac{X - \mu}{\sigma} > \frac{40 - 29}{5}\right] \\ &= P[Z > 2.2] = P[Z \geq 0] - P[0 \leq Z \leq 2.2] = 0.5 - 0.4861 = 0.0139 \end{aligned}$$



Finding Area Under the Normal Curve using Z-table

Example 8:

- a. For what value of 'a', $P[Z \leq a] = 0.95$?
- b. For what value of 'a', $P[Z \geq a] = 0.05$?
- c. For what value of 'a', $P[Z \leq a] = 0.975$?

Solution:

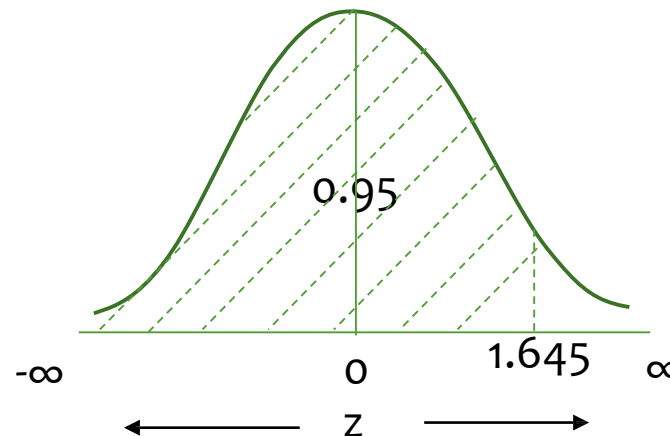
a. $P[Z \leq a] = 0.95$

Or, $P[Z \leq 0] + P[0 < Z \leq a] = 0.95$

Or, $0.5 + P[0 < Z \leq a] = 0.95$

Or, $P[0 < Z \leq a] = 0.95 - 0.5 = 0.45$

For $a = 1.645$, $P[0 < Z \leq a] = 0.45$



Finding Area Under the Normal Curve using Z-table

Solution (contd.):

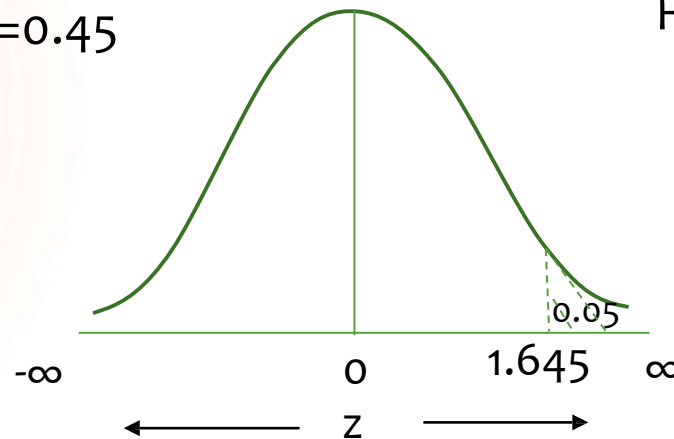
b. $P[Z \geq a] = 0.05$

Or, $P[Z \geq 0] - P[0 < Z \leq a] = 0.05$

Or, $0.5 - P[0 < Z \leq a] = 0.05$

Or, $P[0 < Z \leq a] = 0.5 - 0.05 = 0.45$

For $a = 1.645$, $P[0 < Z \leq a] = 0.45$



Solution (contd.):

c. $P[Z \leq a] = 0.975$

Or, $P[Z \leq 0] + P[0 < Z \leq a] = 0.975$

Or, $0.5 + P[0 < Z \leq a] = 0.975$

Or, $P[0 < Z \leq a] = 0.975 - 0.5 = 0.475$

For $a = 1.96$, $P[0 < Z \leq a] = 0.475$

